ENGINEERING MATHEMATICS

ANALYTIC FUNCTIONS

UNIT- III
ANALYTIC FUNCTIONS

3.1 INTRODUCTION

The theory of functions of a complex variable is the most important in solving a large number of Engineering and Science problems. Many complicated intergrals of real function are solved with the help of a complex variable.

3.1 (a) Complex Variable

$x + iy$ is a complex variable and it is denoted by z.

$(i.e.)z = x + iy \; where \; i = \sqrt{-1}$

3.1 (b) Function of a complex Variable

If $z = x + iy$ and $w = u + iv$ are two complex variables, and if for each value of z in a given region R of complex plane there corresponds one or more values of w is said to be a function z and is denoted by $w = f(z) = f(x + iy) = u(x, y) + iv(x, y)$where $u(x, y)$and $v(x, y)$are real functions of the real variables x and y.

Note:

(i) single-valued function

If for each value of z in R there is correspondingly only one value of w, then w is called a single valued function of z.

Example: $w = z^2, w = \frac{1}{z}$

$w = z^2$					$w = \frac{1}{z}$				
z	1	2	-2	3	z	1	2	-2	3
w	1	4	4	9	w	1	$\frac{1}{2}$	$\frac{1}{-2}$	$\frac{1}{3}$

(ii) Multiple – valued function

If there is more than one value of w corresponding to a given value of z then w is called multiple – valued function.

Example: $w = z^{1/2}$

$w = z^{1/2}$			
z	4	9	1
w	$-2,2$	$3,-3$	$1,-1$

(iii) The distance between two points $z$and z_o is $|z - z_o|$

(iv) The circle C of radius δ with centre at the point $z_o$can be represented by $|z - z_o| = \delta$.

(v) $|z - z_o| < \delta$ represents the interior of the circle excluding its circumference.

(vi) $|z - z_o| \leq \delta$ represents the interior of the circle including its circumference.

(vii) $|z - z_o| > \delta$ represents the exterior of the circle.

(viii) A circle of radius 1 with centre at origin can be represented by $|z| = 1$

3.1 (c) Neighbourhood of a point z_o

Neighbourhood of a point z_o, we mean a sufficiently small circular region [excluding the points on the boundary] with centre at z_o.

$$(i.e.)\ |z - z_o| < \delta$$

Here, δ is an arbitrary small positive number.

3.1 (d) Limit of a Function

Let $f(z)$ be a single valued function defined at all points in some neighbourhood of point z_o.

Then the limit of $f(z)$ as z approaches $z_o is\ w_o$.

$$(i.e.)\ \lim_{z\to z_o} f(z) = w_o$$

3.1 (e) Continuity

If $f(z)$ is said to continuous at $z = z_o$ then

$$\lim_{z\to z_o} f(z) = f(z_o)$$

If two functions are continuous at a point their sum, difference and product are also continuous at that point, their quotient is also continuous at any such point $[dr \neq 0]$

Example: 3.1 State the basic difference between the limit of a function of a real variable and that of a complex variable.

Solution:

In real variable, $x \to x_0$ implies that x approaches x_0 along the X-axis (or) a line parallel to the X-axis.

In complex variables, $z \to z_0$ implies that z approaches z_0 along any path joining the points $z\ and\ z_0$ that lie in the z-plane.

3.1 (f) Differentiability at a point

A function $f(z)$ is said to be differentiable at a point, $z = z_0$ if the limit

$$f(z_0) = \operatorname*{Lt}_{\Delta z\to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \text{ exists.}$$

This limit is called the derivative of $f(z)$at the point $z = z_0$

If $f(z)$ is differentiable at z_0, then $f(z)$is continuous at z_0. This is the necessary condition for differentiability.

Example: 3.2 If $f(z)$ is differentiable at z_0, then show that it is continuous at that point.

Solution:

As $f(z)$ is differentiable at z_0, both $f(z_0)$ and $f'(z_0)$exist finitely.

Now, $\lim_{z\to z_o} |f(z) - f(z_o)| = \lim_{z\to z_o} \frac{f(z)-f(z_o)}{z-z_0}(z - z_0)$

$$= \lim_{z \to z_o} \frac{f(z) - f(z_o)}{z - z_0} \lim_{z \to z_o} (z - z_0)$$

$$= f'(z_0).0 = 0$$

Hence, $\lim_{z \to z_o} f(z) = \lim_{z \to z_o} f(z_o) = f(z_o)$

As $f(z_0)$ is a constant.

This is exactly the statement of continuity of $f(z)$ at z_0.

Example: 3.3 Give an example to show that continuity of a function at a point does not imply the existence of derivative at that point.

Solution:

Consider the function $w = |z|^2 = x^2 + y^2$

This function is continuous at every point in the plane, being a continuous function of two real variables. However, this is not differentiable at any point other than origin.

Example: 3.4 Show that the function $\mathbf{f(z)}$ is discontinuous at $\boldsymbol{z = 0}$, given that $\boldsymbol{f(z) = \frac{2xy^2}{x^2+3y^4}}$, when $\boldsymbol{z \neq 0}$ and $\mathbf{f(0) = 0}$.

Solution:

Given $f(z) = \frac{2xy^2}{x^2+3y^4}$,

Consider $\lim_{z \to z_o} [f(z)] = \lim_{\substack{y=mx \\ x\to 0}} [f(z)] = \lim_{x\to 0} \frac{2x(mx)^2}{x^2+3(mx)^4} = \lim_{x\to 0} \left[\frac{2m^2x}{1+3m^4x^2}\right] = 0$

$$\lim_{\substack{y^2=x \\ x\to 0}} [f(z)] = \lim_{x\to 0} \frac{2x^2}{x^2 + 3x^2} = \lim_{x\to 0} \frac{2x^2}{4x^2} = \frac{2}{4} = \frac{1}{2} \neq 0$$

$\therefore f(z)$ is discontinuous

Example: 3.5 Show that the function $\boldsymbol{f(z)}$ is discontinuous at the origin $(\boldsymbol{z = 0})$, given that

$\boldsymbol{f(z) = \frac{xy(x-2y)}{x^3+y^3}}$**, when $\boldsymbol{z \neq 0}$**

$\mathbf{= 0}$ **, when $\boldsymbol{z = 0}$**

Solution:

Consider $\lim_{z \to z_o} [f(z)] = \lim_{\substack{y=mx \\ x\to 0}} [f(z)] = \lim_{x\to 0} \frac{x(mx)(x-2(mx))}{x^3+(mx)^3}$

$$= \lim_{x\to 0} \frac{m(1-2m)x^3}{(1+m^3)x^3} = \frac{m(1-2m)}{1+m^3}$$

Thus $\lim_{z\to 0} f(z)$ depends on the value of m and hence does not take a unique value.

$\therefore \lim_{z\to 0} f(z)$does not exist.

$\therefore f(z)$ is discontinuous at the origin.

3.2 ANALYTIC FUNCTIONS – NECESSARY AND SUFFICIENT CONDITIONS FOR ANALYTICITY IN CARTESIAN AND POLAR CO-ORDINATES

Analytic [or] Holomorphic [or] Regular function

A function is said to be analytic at a point if its derivative exists not only at that point but also in some neighbourhood of that point.

Entire Function: [Integral function]

A function which is analytic everywhere in the finite plane is called an entire function.

An entire function is analytic everywhere except at $z = \infty$.

Example: $e^z, \sin z, \cos z, sinhz, \cosh z$

3.2 (i) The necessary condition for $f = (z)$ to be analytic. [Cauchy – Riemann Equations]

The necessary conditions for a complex function $f = (\text{z}) = u(x,y) + iv(x,y)$ to be analytic in a region R are $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$ $i.e., u_x = v_y$ and $v_x = -u_y$

[OR]

Derive C – R equations as necessary conditions for a function $\mathbf{w = f(z)}$ to be analytic.

Proof:

Let $f(z) = u(x,y) + iv(x,y)$ be an analytic function at the point z in a region R. Since $f(z)$ is analytic, its derivative $f^{'}(z)$ exists in R

$$f'(z) = \text{Lt}\frac{f(z+\Delta z)-f(z)}{\Delta_z}$$

Let $z = x + iy$

$$\Rightarrow \Delta z = \Delta_x + i\Delta_y$$

$$z + \Delta_z = (x + \Delta_x) + i\left(y + \Delta_y\right)$$

$$f(z) = u(x,y) + iv(x,y)$$

$$f(z + \Delta z) = u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y$$

$$f(z + \Delta z) - f(z) = u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - [u(x,y) + iv(x,y)]$$

$$= [u(x + \Delta x, y + \Delta y) - u(x,y)] + i[v(x + \Delta x, y + \Delta y) - v(x,y)]$$

$$f'(z) = \underset{\Delta z \to 0}{\text{Lt}} \frac{f(z+\Delta z)-f(z)}{\Delta z}$$

$$= \underset{\Delta z \to 0}{\text{Lt}} \frac{u(x+\Delta x,y+\Delta y)-u(x,y)]+i[v(x+\Delta x,y+\Delta y)-v(x,y)]}{\Delta x+i\Delta y}$$

Case (i)

If $\Delta z \to 0$,firsts we assume that $\Delta y = 0$ and $\Delta x \to 0$.

$$\therefore\ f'(z) = \underset{\Delta x \to 0}{\text{Lt}} \frac{[u(x+\Delta x,y)-u(x,y)]+i[v(x+\Delta x,y)-v(x,y)]}{\Delta x}$$

$$= \underset{\Delta x \to 0}{\text{Lt}} \frac{u(x+\Delta x,y)-u(x,y)}{\Delta x} + \underset{\Delta x \to 0}{\text{Lt}} \frac{v(x+\Delta x,y)-v(x,y)}{\Delta x}$$

$$= \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \qquad \ldots(1)$$

Case (ii)

If $\Delta z \to 0$ Now, we assume that $\Delta x = 0$ and $\Delta y \to 0$

$$\therefore\ f'(z) = \underset{\Delta y \to 0}{\text{Lt}} \frac{[u(x,y+\Delta y)-u(x,y)]+i[v(x,y+\Delta y)-v(x,y)]}{i\Delta y}$$

$$= \frac{1}{i} \underset{\Delta y \to 0}{\text{Lt}} \frac{u(x,y+\Delta y)-u(x,y)}{\Delta y} + \underset{\Delta y \to 0}{\text{Lt}} \frac{v(x,y+\Delta y)-v(x,y)}{\Delta y}$$

$$= \frac{1}{i}\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$$

$$= -i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \qquad \ldots(2)$$

From (1) and (2), we get

$$\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = -\,i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$$

Equating the real and imaginary parts we get

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial v}{\partial x} = -\frac{\partial v}{\partial y}$$

$(i.e.)\ u_x = v_y, \quad v_x = -u_y$

The above equations are known as Cauchy – Riemann equations or C-R equations.

Note: (i) The above conditions are not sufficient for $f(z)$ to be analytic. The sufficient conditions are given in the next theorem.

(ii) Sufficient conditions for $f(z)$ to be analytic.

If the partial derivatives u_x, u_y, v_x and v_y are all continuous in D and $u_x = v_y$ and $u_y = -v_x$, then the function $f(z)$ is analytic in a domain D.

(ii) Polar form of C-R equations

In Cartesian co-ordinates any point z is $z = x + iy$.

In polar co-ordinates, $z = re^{i\theta}$ where r is the modulus and θ is the argument.

Theorem: If $f(z) = u(r,\theta) + iv(r,\theta)$ is differentiable at $z = re^{i\theta}$, then $u_r = \frac{1}{r}v_\theta, v_r = -\frac{1}{r}u_\theta$

(OR) $\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \frac{\partial v}{\partial r} = \frac{-1}{r}\frac{\partial u}{\partial \theta}$

Proof:

Let $z = re^{i\theta}$ and $w = f(z) = u + iv$

$(\,i.e.)\ u + iv = f(z) = f(re^{i\theta})$

Diff. p.w. r. to r, we get

$$\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r} = f'(re^{i\theta})\, e^{i\theta} \qquad \ldots(1)$$

Diff. p.w. r. to θ, we get

$$\frac{\partial u}{\partial \theta} + i\frac{\partial v}{\partial \theta} = f'(re^{i\theta})\, e^{i\theta} \qquad \ldots(2)$$

$$= ri\left[f'(re^{i\theta})\, e^{i\theta}\right]$$

$$= ri\left[\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right] \quad \text{by} \quad (1)$$

$$= ri\frac{\partial u}{\partial r} - r\frac{\partial v}{\partial r}$$

Equating the real and imaginary parts, we get

$$\frac{\partial u}{\partial \theta} = -\,i\frac{\partial v}{\partial r}, \quad \frac{\partial v}{\partial \theta} = r\frac{\partial u}{\partial r}$$

$$(i.e.)\ \frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = \frac{-1}{r}\frac{\partial v}{\partial \theta}$$

Problems based on Analytic functions – necessary conditions Cauchy – Riemann equations

Example: 3.6 Show that the function $\mathrm{f}(\mathrm{z}) = xy + iy$ is continuous everywhere but not differentiable anywhere.

Solution:

Given $f(z) = xy + iy$

$(i.e.)\ \ u = xy, v = y$

x and y are continuous everywhere and consequently $u(x, y) = xy$ and $v(x, y) = y$ are continuous everywhere.

Thus $f(z)$ is continuous everywhere.

But

$u = xy$	$v = y$
$u_x = y$	$v_x = 0$
$u_y = x$	$v_y = 1$
$u_x \neq v_y$	$u_y \neq -v_x$

C−R equations are not satisfied.

Hence, $f(z)$ is not differentiable anywhere though it is continuous everywhere .

Example: 3.7 Show that the function $f(z) = \bar{z}$ is nowhere differentiable.

Given $f(z) = \bar{z} = x - iy$

Solution:

$u = x$	$v = -y$
$\frac{\partial u}{\partial x} = 1$	$\frac{\partial v}{\partial x} = 0$
$\frac{\partial u}{\partial y} = 0$	$\frac{\partial v}{\partial y} = -1$

$$\therefore u_x \neq v_y$$

C−R equations are not satisfied anywhere.

Hence, $f(z) = \bar{z}$ is not differentiable anywhere (or) nowhere differentiable.

Example: 3.8 Show that $\mathrm{f}(\mathrm{z}) = |\mathrm{z}|^2$ is differentiable at $z = 0$ but not analytic at $z = 0$.

Solution:

Let $z = x + iy$

$\bar{z} = x - iy$

$|\mathrm{z}|^2 = \mathrm{z}\,\bar{z} = x^2 + y^2$

$(i.e.)\ f(z) = |z|^2 = (x^2 + y^2) + i0$

$u = x^2 + y^2$	v= 0
$u_x = 2x$ $u_y = 2y$	$v_x = 0$ $v_y = 0$

So, the C−R equations $u_x = v_y$ and $u_y = -v_x$ are not satisfied everywhere except at $z = 0$.

So, f(z) may be differentiable only at $z = 0$.

Now, $u_x = 2x$, $u_y = 2y$,$v_x = 0$ and $v_y = 0$ are continuous everywhere and in particular at (0,0).

Hence, the sufficient conditions for differentiability are satisfied by $f(z)$ at $z = 0$.

So, $f(z)$ is differentiable at $z = 0$ only and is not analytic there.

Inverse function

Let $w = f(z)$ be a function of z and $z = F(w)$ be its inverse function.

Then the function $w = f(z)$ will cease to be analytic at $\frac{dz}{dw} = 0$ and $z = F(w)$ will be so, at point where $\frac{dw}{dz} = 0$.

Example: 3.9 Show that $f(z) = log\ z$ analytic everywhere except at the origin and find its derivatives.

Solution:

$$\text{Let } z = re^{i\theta}$$

$$f(z) = log\ z$$

$$= \log\left(re^{i\theta}\right) = log r + \log\left(e^{i\theta}\right) = log r + i\theta$$

But, at the origin, $r = 0$. Thus, at the origin,

$$f(z) = log 0 + i\theta = -\infty + i\theta$$

Note : $e^{-\infty} = 0$

$\log e^{-\infty} = \log 0\,; -\infty = log 0$

So, $f(z)$is not defined at the origin and hence is not differentiable there.

At points other than the origin, we have

$u(r,\theta) = log\ r$	$v(r,\theta) = \theta$
$u_r = \frac{1}{r}$ $u_\theta = 0$	$v_r = 0$ $v_\theta = 1$

So, $logz$ satisfies the C−R equations.

Further $\frac{1}{r}$ is not continuous at $z = 0$.

So, $u_r, u_\theta, v_r, v_\theta$ are continuous everywhere except at $z = 0$. Thus log z satisfies all the sufficient conditions for the existence of the derivative except at the origin. The derivative is

$$f'(z) = \frac{u_r + iv_r}{e^{i\theta}} = \frac{\left(\frac{1}{r}\right) + i(0)}{e^{i\theta}} = \frac{1}{re^{i\theta}} = \frac{1}{z}$$

Note: $f(z) = u + iv \Rightarrow f(re^{i\theta}) = u + iv$

Differentiate w.r.to ‘r’, we get

$$(i.e.)\ e^{i\theta} f'(re^{i\theta}) = \frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}$$

Example: 3.10 Check whether $w = \bar{z}$ is analytics everywhere.

Solution:

Let $w = f(z) = \bar{z}$

u+$iv = x - iy$

$u = x$	$v = -y$
$u_x = 1$ $u_y = 0$	$v_x = 0$ $v_y = -1$

$u_x \neq v_y$ at any point p(x,y)

Hence, C−R equations are not satisfied.

∴The function $f(z)$ is nowhere analytic.

Example: 3.11 Test the analyticity of the function $w = sin\ z$.

Solution:

Let $w = f(z) = sinz$

$u + iv = \sin(x + iy)$

$u + iv = \sin x \cos iy + \cos x \sin iy$

$u + iv = \sin x \cosh y + i \cos x \sin hy$

Equating real and imaginary parts, we get

$u = \sin x \cosh y$	$v = \cos x \sinh y$
$u_x = \cos x \cosh y$ $u_y = \sin x \sinh y$	$v_x = -\sin x \sinh y$ $v_y = \cos x \cosh y$

$\therefore u_x = v_y$ and $u_y = -v_x$

C −R equations are satisfied.

Also the four partial derivatives are continuous.

Hence, the function is analytic.

Example: 3.12 Determine whether the function $2xy + i(x^2 - y^2)$ is analytic or not.

Solution:

Let $f(z) = 2xy + i(x^2 - y^2)$

$(i.e.)$

$u = 2xy$	$v = x^2 - y^2$
$\frac{\partial u}{\partial x} = 2y$ $\frac{\partial u}{\partial y} = 2x$	$\frac{\partial v}{\partial x} = 2x$ $\frac{\partial v}{\partial y} = -2y$

$u_x \neq v_y$ and $u_y \neq -v_x$

C$-$R equations are not satisfied.

Hence, $f(z)$ is not an analytic function.

Example: 3.13 Prove that $f(z) = \cosh z$ is an analytic function and find its derivative.

Solution:

Given $f(z) = \cosh z = \cos(iz) = \cos[i(x+iy)]$

$= \cos(ix - y) = \cos ix \cos y + \sin(ix) \sin y$

$u + iv = \cosh x \cos y + i \sinh x \sin y$

$u = \cosh x \cos y$	$v = \sinh x \sin y$
$u_x = \sinh x \cos y$ $u_y = -\cosh x \sin y$	$v_x = \cosh x \sin y$ $v_y = \sinh x \cos y$

$\therefore u_x, u_y, v_x$ and v_y exist and are continuous.

$u_x = v_y$ and $u_y = -v_x$

C$-$R equations are satisfied.

$\therefore f(z)$ is analytic everywhere.

Now, $f'(z) = u_x + iv_x$

$= \sinh x \cos y + i \cosh x \sin y$

$= \sinh(x+iy) = \sinh z$

Example: 3.14 If $w = f(z)$ is analytic, prove that $\frac{dw}{dz} = \frac{\partial w}{\partial x} = -i\frac{\partial w}{\partial y}$ where $z = x + iy$, and prove that $\frac{\partial^2 w}{\partial z \partial \bar{z}} = 0$. [Anna, Nov 2001]

Solution:

Let $w = u(x,y) + iv(x,y)$

As $f(z)$ is analytic, we have $u_x = v_y, u_y = -v_x$

Now, $\frac{dw}{dz} = f'(z) = u_x + iv_x = v_y - iu_y = i(u_y + iv_y)$

$= \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = -i\left[\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right]$

$= \frac{\partial}{\partial x}(u + iv) = -i\frac{\partial}{\partial y}(u + iv)$

$= \frac{\partial w}{\partial x} = -i\frac{\partial w}{\partial y}$

We know that, $\frac{\partial w}{\partial z} = 0$

$\therefore \frac{\partial^2 w}{\partial z \partial \bar{z}} = 0$

Also $\frac{\partial^2 w}{\partial \bar{z} \partial z} = 0$

Example: 3.15 Prove that every analytic function $w = u(x,y) + iv(x,y)$ can be expressed as a function of z alone.

Proof:

Let $z = x + iy$ and $\bar{z} = x - iy$

$$x = \frac{z+\bar{z}}{2} \quad \text{and} \quad y = \frac{z+\bar{z}}{2i}$$

Hence, u and v and also w may be considered as a function of z and $\bar{z}$

Consider $\frac{\partial w}{\partial \bar{z}} = \frac{\partial u}{\partial \bar{z}} + i\frac{\partial v}{\partial \bar{z}}$

$$= \left(\frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial \bar{z}} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial \bar{z}}\right) + \left(\frac{\partial v}{\partial x}\frac{\partial x}{\partial \bar{z}} + \frac{\partial v}{\partial y}\frac{\partial y}{\partial \bar{z}}\right)$$

$$= \left(\frac{1}{2}u_x - \frac{1}{2i}u_y\right) + i\left(\frac{1}{2}v_x - \frac{1}{2i}v_y\right)$$

$$= \frac{1}{2}(u_x - v_y) + \frac{i}{2}(u_y + v_x)$$

$= 0$ by C−R equations as w is analytic.

This means that w is independent of $\bar{z}$

$(i.e.)$ w is a function of z alone.

This means that if $w = u(x,y) + iv(x,y)$ is analytic, it can be rewritten as a function of $(x + iy)$.

Equivalently a function of $\bar{z}$ cannot be an analytic function of z.

Example: 3.16 Find the constants a, b, c if $\mathbf{f(z) = (x + ay) + i(bx + cy)}$ is analytic.

Solution:

$$f(z) = u(x,y) + iv(x,y)$$

$$= (x + ay) + i(bx + cy)$$

$u = x + ay$	$v = bx + cy$
$u_x = 1$ $u_y = a$	$v_x = b$ $v_y = c$

Given $f(z)$ is analytic

$\Rightarrow u_x = v_y$ and $u_y = -v_x$

$1 = c$ and $a = -b$

Example: 3.17 Examine whether the following function is analytic or not $\mathbf{f(z) = e^{-x}(\cos y - i \sin y)}$.

Solution:

Given $f(z) = e^{-x}(\cos y - i \sin y)$

$\Rightarrow u + iv = e^{-x}\cos y - ie^{-x}\sin y$

$u = e^{-x}\cos y$	$v = -e^{-x}\sin y$
$u_x = -e^{-x}\cos y$ $u_y = -e^{-x}\sin y$	$v_x = e^{-x}\sin y$ $v_y = -e^{-x}\cos y$

Here, $u_x = v_y$ and $u_y = -v_x$

$\Rightarrow$ C−R equations are satisfied

$\Rightarrow f(z)$ is analytic.

Example: 3.18 Test whether the function $\mathbf{f(z) = \frac{1}{2}\log(x^2 + y^2 + tan^{-1}\left(\frac{y}{x}\right)}$ is analytic or not.

Solution:

Given $f(z) = \frac{1}{2}\log(x^2 + y^2 + i\tan^{-1}\left(\frac{y}{x}\right)$

$(i.e.) u + iv = \frac{1}{2}\log(x^2 + y^2 + i\tan^{-1}\left(\frac{y}{x}\right)$

$u = \frac{1}{2}\log(x^2 + y^2)$	$v = \tan^{-1}\left(\frac{y}{x}\right)$
$u_x = \frac{1}{2}\frac{1}{x^2+y^2}(2x)$ $= \frac{x}{x^2+y^2}$ $u_y = \frac{1}{2}\frac{1}{x^2+y^2}(2y)$ $= \frac{y}{x^2+y^2}$	$v_x = \frac{1}{1+\frac{y^2}{x^2}}\left[-\frac{y}{x^2}\right]$ $= \frac{-y}{x^2+y^2}$ $v_y = \frac{1}{1+\frac{y^2}{x^2}}\left[\frac{1}{x}\right]$ $= \frac{x}{x^2+y^2}$

Here, $u_x = v_y$ and $u_y = -v_x$

$\Rightarrow$ C−R equations are satisfied

$\Rightarrow f(z)$ is analytic.

Example: 3.19 Find where each of the following functions ceases to be analytic.

(i) $\frac{z}{(z^2-1)}$ **(ii)** $\frac{z+i}{(z-i)^2}$

Solution:

(i) Let $f(z) = \frac{z}{(z^2-1)}$

$$f'(z) = \frac{(z^2-1)(1)-z(2z)}{(z^2-1)^2} = \frac{-(z^2+1)}{(z^2-1)^2}$$

$f(z)$ is not analytic, where $f'(z)$ does not exist.

$(i.e.)\ f'(z) \to \infty$

$(i.e.)(z^2-1)^2 = 0$

$(i.e.)\quad z^2 - 1 = 0$

$z = 1$

$z = \pm 1$

$\therefore f(z)$ is not analytic at the points $z = \pm 1$

(ii) Let $f(z) = \frac{z+i}{(z-i)^2}$

$$f'(z) = \frac{(z-i)^2(1)(z+i)[2(z-i)]}{(z-i)^4} = \frac{(z+3i)}{(z-i)^3}$$

$$f'(z) \to \infty, at\ z = i$$

$\therefore f(z)$ is not analytic at $z = i$.

Exercise: 3.1

1. Examine the following function are analytic or not

1. $f(z) = e^x(\cos y + i \sin y)$ [**Ans:** analytic]
2. $f(z) = e^x(\cos y - i \sin y)$ [**Ans:** not analytic]
3. $f(z) = z^3 + z$ [**Ans:** analytic]
4. $f(z) = \sin x \cos y + i \cos x \sinh y$ [**Ans:** analytic]
5. $f(z) = (x^2 - y^2 + 2xy) + i(x^2 - y^2 - 2xy)$ [**Ans:** not analytic]
6. $f(z) = 2xy + i(x^2 - y^2)$ [**Ans:** not analytic]
7. $f(z) = \cosh z$ [**Ans:** analytic]
8. $f(z) = y$ [**Ans:** not analytic]
9. $f(z) = (x^2 - y^2 - 2xy) + i(x^2 - y^2 + 2xy)$ [**Ans:** analytic]
10. $f(z) = \frac{x-iy}{x^2+y^2}$ [**Ans:** analytic]

2. For what values of z, the function ceases to be analytic.

1. $\frac{1}{z^2-4}$ [**Ans:** $z = \pm 1$]
2. $\frac{z^2-4}{z^2+1}$ [**Ans:** $z = \pm 1$]

3. Verify C-R equations for the following functions.

1. $f(z) = ze^z$
2. $f(z) = lz + m$
3. $f(z) = \cos z$

4. Prove that the following functions are nowhere differentiable.

1. $f(z) = e^x(\cos y - i \sin y)$
2. $f(z) = |z|$
3. $f(z) = z - \bar{z}$

5. Find the constants a, b, c so that the following are differentiable at every point.

1. $f(z) = x + ay - i(bx + cy)$ [**Ans.** $a = b, c = -1$]
2. $f(z) = ax^2 - by^2 + i\, cxy$ [**Ans.** $a = b = \frac{c}{2}$]

3.3 PROPERTIES – HARMONIC CONJUGATES

3.3 (a) Laplace equation

$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} = 0$ is known as Laplace equation in two dimensions.

3.3 (b) Laplacian Operator

$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is called the Laplacian operator and is denoted by ∇^2.

Note: (i) $\frac{\partial^2\varphi}{\partial x^2}+\frac{\partial^2\varphi}{\partial y^2}+\frac{\partial^2\varphi}{\partial z^2}=0$ is known as Laplace equation in three dimensions.

Note: (ii) The Laplace equation in polar coordinates is defined as

$$\frac{\partial^2\varphi}{\partial r^2}+\frac{1}{r}\frac{\partial\varphi}{\partial r}+\frac{1}{r^2}\frac{\partial^2\varphi}{\partial\theta^2}=0$$

Properties of Analytic Functions

Property: 1 Prove that the real and imaginary parts of an analytic function are harmonic functions.

Proof:

Let $f(z)=u+iv$ be an analytic function

$u_x=v_y\ldots(1)$ and $u_y=-v_x\quad\ldots(2)$ by C−R

Differentiate (1) & (2) p.w.r. to x, we get

$u_{xx}=v_{xy}\ldots(3)$ and $u_{xy}=-v_{xx}\quad\ldots(4)$

Differentiate (1) & (2) p.w.r. to x, we get

$u_{yx}=v_{yy}\ldots(5)$ and $u_{yy}=-v_{yx}\quad\ldots(6)$

(3) +(6) ⇒ $u_{xx}+u_{yy}=0\quad[\because v_{xy}=v_{yx}]$

(5) – (4) ⇒ $v_{xx}+v_{yy}=0\quad[\because u_{xy}=u_{yx}]$

$\therefore$ u and v satisfy the Laplace equation.

3.3 (c) Harmonic function (or) [Potential function]

A real function of two real variables x and y that possesses continuous second order partial derivatives and that satisfies Laplace equation is called a harmonic function.

Note: A harmonic function is also known as a potential function.

3.3 (d) Conjugate harmonic function

If u and v are harmonic functions such that $u+iv$ is analytic, then each is called the conjugate harmonic function of the other.

Property: 2 If w $=u(x,y)+iv(x,y)$ is an analytic function the curves of the family $u(x,y)=c_1$ and the curves of the family $v(x,y)=c_2$ cut orthogonally, where c_1 and c_2 are varying constants.

Proof:

Let $f(z)=u+iv$ be an analytic function

$\Rightarrow u_x=v_y\ldots(1)$ and $u_y=-v_x\quad\ldots(2)$ by C−R

Given $u=c_1$ and $v=c_2$

Differentiate p.w.r. to x, we get

$$u_x+u_y\frac{dy}{dx}=0 \text{ and } \quad v_x+v_y\frac{dy}{dx}=0$$

$$\Rightarrow\frac{dy}{dx}=\frac{-u_x}{u_y}\text{ and }\frac{dy}{dx}=\frac{-v_x}{v_y}$$

$$\Rightarrow m_1=\frac{-u_x}{u_y}\qquad\Rightarrow m_2=\frac{-v_x}{v_y}$$

$$m_1.m_2 = \left(\frac{-u_x}{u_y}\right)\left(\frac{-v_x}{v_y}\right) = \left(\frac{u_x}{u_y}\right)\left(\frac{u_y}{u_x}\right) = -1 \text{ by (1) and (2)}$$

Hence, the family of curves form an orthogonal system.

Property: 3 An analytic function with constant modulus is constant.

Proof:

Let $f(z) = u + iv$ be an analytic function.

$\Rightarrow u_x = v_y \ldots(1)$ and $u_y = -v_x \ldots(2)$ by C−R

Given $|f(z)| = \sqrt{u^2 + v^2} = c \neq 0$

$\Rightarrow |f(z)| = u^2 + v^2 = c^2$ (say)

$(i.e)\ u^2 + v^2 = c^2 \quad \ldots(3)$

Differentiate (3) p.w.r. to *x* and *y*; we get

$2uu_x + 2vv_x = 0 \Rightarrow uu_x + vv_x = 0 \quad \ldots(4)$

$2uu_y + 2vv_y = 0 \Rightarrow uu_y + vv_y = 0 \quad \ldots(5)$

$(4) \times u \quad \Rightarrow u^2u_x + uv\, v_x = 0 \quad \ldots(6)$

$(5) \times v \quad \Rightarrow uv\, u_y + v^2v_y = 0 \quad \ldots(7)$

$(6)+(7) \quad \Rightarrow u^2u_x + v^2v_y + uv\,[v_x + u_y] = 0$

$\Rightarrow u^2u_x + v^2\, u_x + uv\,[-u_y + u_y] = 0$ by (1) & (2)

$\Rightarrow (u^2+v^2)\, u_x = 0$

$\Rightarrow u_x = 0$

Similarly, we get $v_x = 0$

We know that $f'(z) = u_x + v_x = 0 + i0 = 0$

Integrating w.r.to z, we get, $f(z) = c$ [Constant]

Property: 4 An analytic function whose real part is constant must itself be a constant. [A.U M/J 2016]

Proof :

Let $f(z) = u + iv$ be an analytic function.

$\Rightarrow u_x = v_y \ldots(1)$ and $u_y = -v_x \ldots(2)$ by C−R

Given $u = c$ [Constant]

$\Rightarrow u_x = 0, \quad u_y = 0$

$\Rightarrow u_x = 0, \quad v_x = 0$ by (2)

We know that $f'(z) = u_x + iv_x = 0 + i0 = 0$

Integrating w.r.to z, we get $f(z) = c$ [Constant]

Property: 5 Prove that an analytic function with constant imaginary part is constant.

Proof:

Let $f(z) = u + iv$ be an analytic function.

$\Rightarrow u_x = v_y \ldots(1)$ and $u_y = -v_x \ldots(2)$ by C−R

Given $v = c$ [Constant]

$\Rightarrow v_x = 0, \qquad v_y = 0$

We know that $f^{'}(z) = u_x + iv_x$

$= v_y + iv_x$ by (1) $= 0 + i0$

$\Rightarrow f^{'}(z) = 0$

Integrating w.r.to z, we get $f(z) = c$ [Constant]

Property: 6 If $\mathbf{f(z)}$ and $\overline{\mathbf{f(z)}}$ are analytic in a region D, then show that $\mathbf{f(z)}$ is constant in that region D.

Proof:

Let $f(z) = u(x,y) + iv\,(x,y)$ be an analytic function.

$\overline{f(z)} = u(x,y) - iv(x,y) = u(x,y) + i[-v(x,y)]$

Since, $f(z)$ is analytic in D, we get $u_x = v_y$ and $u_y = -v_x$

Since, $\overline{f(z)}$ is analytic in D, we have $u_x = -v_y$ and $u_y = v_x$

Adding, we get $u_x = 0$ and $u_y = 0$ and hence, $v_x = v_y = 0$

$\therefore\ f(z) = u_x + iv_x = 0 + i0 = 0$

$\therefore\ f(z)$ is constant in D.

Problems based on properties

Theorem: 1 If $\mathbf{f(z)} = \boldsymbol{u + iv}$ is a regular function of z in a domain D, then $\boldsymbol{\nabla^2 \left|f(z)\right|^2 = 4\left|f'(z)\right|^2}$

Solution:

Given $f(z) = u + iv$

$\Rightarrow\ |f(z)| = \sqrt{u^2 + v^2}$

$\Rightarrow |f(z)|^2 = u^2 + v^2$

$\Rightarrow \nabla^2|f(z)|^2 = \nabla^2(u^2 + v^2)$

$= \nabla^2(u^2) + \nabla^2(v^2) \qquad \ldots(1)$

$$\nabla^2(u^2) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)u^2 += \frac{\partial^2(u^2)}{\partial x^2} + \frac{\partial^2(u^2)}{\partial y^2} \qquad \ldots(2)$$

$$\frac{\partial^2}{\partial x^2}(u^2) = \frac{\partial}{\partial x}\left[2u\frac{\partial u}{\partial x}\right] = 2\left[u\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x}\frac{\partial u}{\partial x}\right] = 2u\frac{\partial^2 u}{\partial x^2} + 2\left(\frac{\partial u}{\partial x}\right)^2$$

Similarly, $\frac{\partial^2}{\partial y^2}(u^2) = 2u\frac{\partial^2 u}{\partial y^2} + 2\left(\frac{\partial u}{\partial y}\right)^2$

$$(2)\Rightarrow\ \nabla^2(u^2) = 2u\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + 2\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2\right]$$

$$= 0 + 2\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2\right] \qquad [\therefore u \text{ is harmonic}]$$

$\nabla^2(u^2) = 2u_x^2 + 2u_y^2$

Similarly, $\nabla^2(v^2) = 2v_x^2 + 2v_y^2$

$(1)\Rightarrow\ \nabla^2|f(z)|^2 = 2\left[u_x^2 + u_y^2 + v_x^2 + v_y^2\right]$

$= 2[u_x^2 + (-v_x)^2 + v_x^2 + u_x^2] \qquad [\because u_x = v_y; u_y = -v_x]$

$$= 4[u_x^2 + v_x^2]$$

$$(i.e.)\nabla^2|f(z)|^2 = 4|f'(z)|^2$$

Note : $f(z) = u + iv$; $f'(z) = u_x + iv_x$;

(or) $f'(z) = v_y + iu_y$; $|f'(z)| = \sqrt{u_x^2 + v_x^2}$; $|f'(z)|^2 = u_x^2 + v_x^2$

Theorem: 2 If $\mathbf{f(z) = u + iv}$ is a regular function of z in a domain D, then $\boldsymbol{\nabla^2} \log |\mathbf{f(z)}| = 0$ if $\mathbf{f(z)\, f'(z)} \neq 0$ in D. i.e., $\log |\mathbf{f(z)}|$ is harmonic in D.

Solution:

Given $f(z) = u + iv$

$$|f(z)| = \sqrt{u^2 + v^2}$$

$$\log |f(z)| = \frac{1}{2} \log (u^2 + v^2)$$

$$\nabla^2 \log|f(z)| = \frac{1}{2} \nabla^2 \log (u^2 + v^2) = \frac{1}{2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\log(u^2 + v^2)$$

$$= \frac{1}{2}\frac{\partial^2}{\partial x^2}[log(u^2 + v^2)] + \frac{1}{2}\frac{\partial^2}{\partial y^2}[log(u^2 + v^2)] \qquad \ldots(1)$$

$$\frac{1}{2}\frac{\partial^2}{\partial x^2}[log(u^2 + v^2)] = \frac{1}{2}\frac{\partial}{\partial x}\left[\frac{1}{u^2+v^2}\left(2u\frac{\partial u}{\partial x} + 2v\frac{\partial v}{\partial x}\right)\right] = \frac{\partial}{\partial x}\left[\frac{uu_x+vv_x}{u^2+v^2}\right]$$

$$= \frac{(u^2+v^2)[uu_{xx}+u_xu_x+vv_{xx}+v_xv_x]-(uu_x+vv_x)(2uu_x+2vv_x)}{(u^2+v^2)^2}$$

$$= \frac{(u^2+v^2)[uu_{xx}+vv_{xx}+u_x^2+v_x^2]-2(uu_x+vv_x)^2}{(u^2+v^2)^2}$$

Similarly, $\frac{1}{2}\frac{\partial^2}{\partial y^2}[log(u^2 + v^2)] = \frac{(u^2+v^2)[uu_{yy}+vv_{yy}+u_y^2+v_y^2]-2(uu_y+vv_y)^2}{(u^2+v^2)^2}$

(1)$\Rightarrow$ $\nabla^2\log|f(z)| = \frac{(u^2+v^2)[u(u_{xx}+u_{yy})+\mathrm{v}(v_{xx}+v_{yy}) +(u_x^2+u_y^2) +(v_x^2+v_y^2)]-2[uu_x+vv_x]^2-2[uu_y+vv_y]^2}{(u^2+v^2)^2}$

$$= \frac{(u^2+v^2)[u(0)+(u_x^2+v_x^2)+u_y^2+v_y^2)--2[u^2u_x^2+v^2v_x^2+2uv\ u_xv_x+u^2\ u_y^2+v^2v_y^2+2uv\ u_y\,v_y]}{(u^2+v^2)^2}$$

$$[\because u_{xx} + u_{yy} = 0, v_{xx} + vv_{yy} = 0]$$

$$= \frac{(u^2+v^2)[|f'(z)^2|+|f'(z)^2|-2[u^2(u_x^2+ u_y^2)+v^2(v_x^2+v_y^2)+2uv(u_x\ v_x+u_y\,v_y]}{(u^2+v^2)^2}$$

$[\because f'(z) = u + iv, |f'(z)| = u_x + iv_x$ (or) $f'(z) = v_y - iu_{y,}|f'(z)|^2 = u_x^2 + v_x^2$

(or) $|f'(z)|^2 = u_y^2 + v_y^2$

$$= \frac{2(u^2+v^2)[|f'(z)^2|-2[u^2|f'(z)|^2+v^2|f'(z)^2|+2uv(0)]}{(u^2+v^2)^2}$$

$$[\because u_x = v_{y,}\ u_y = -v_x]$$

$$\Rightarrow u_x\ v_x + u_yv_y = 0$$

$$\Rightarrow u_x^2 + u_y^2 = u_x^2 + v_x^2 = |f'(z)|^2$$

$$\Rightarrow v_x^2 + v_y^2 = u_y^2 + v_y^2 = |f'(z)|^2$$

$$= \frac{2(u^2+v^2)|f(z)|^2-2(u^2+v^2)|f'(z)|^2}{(u^2+v^2)^2}$$

$$(i.e.)\ \nabla^2\log|f(z)| = 0$$

Theorem: 3 If $f(z) = u + iv$ is a regular function of z in a domain D, then

$$\nabla^2(u^p) = p(p-1)\, u^{p-2}|f'(z)|^2$$

Solution:

$$\nabla^2(u^p) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)(u^p)$$

$$= \frac{\partial^2}{\partial x^2}(u^p) + \frac{\partial^2}{\partial y^2}(u^p)$$

$$\frac{\partial^2}{\partial x^2}(u^p) = \frac{\partial}{\partial x}\left[pu^{p-1}\frac{\partial u}{\partial x}\right] = pu^{p-1}u_{xx} + p(p-1)u^{p-2}(u_x)^2$$

Similarly, $\frac{\partial^2}{\partial y^2}(u^p) = pu^{p-1}u_{yy} + p(p-1)u^{p-2}(u_y)^2$

$$(1) \Rightarrow \nabla^2(u^p) = pu^{p-1}(u_{xx} + u_{yy}) + p(p-1)u^{p-2}[u_x^2 + u_y^2]$$

$$= pu^{p-1}(0) + p(p-1)u^{p-2}|f'(z)|^2$$

$$[\because u_{xx} + u_{yy} = 0, f(z) = u + iv, f'(z) = u_x + iv_x, \ |f'(z)|^2 = u_x^2 + u_y^2$$

$$\therefore \nabla^2(u^p) = p(p-1)\, u^{p-2}|f'(z)|^2$$

Theorem: 4 If $f(z) = u + iv$ is a regular function of z, then $\nabla^2|f(z)|^p = p^2|f(z)|^{p-2}|f'(z)|^2$.

Solution:

Let $f(z) = u + iv$

$$|f(z)| = \sqrt{u^2 + v^2} \qquad \ldots(a)$$

$$|f(z)|^p = (u^2 + v^2)^{p/2} \qquad \ldots(b)$$

$$\nabla^2|f(z)|^p = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)(u^2 + v^2)^{p/2}$$

$$= \frac{\partial^2}{\partial x^2}(u^2 + v^2)^{p/2} + \frac{\partial^2}{\partial y^2}(u^2 + v^2)^{p/2}$$

$$\frac{\partial^2}{\partial x^2}(u^2 + v^2)^{p/2} = \frac{\partial}{\partial x}\left[\frac{p}{2}(u^2 + v^2)^{\frac{p}{2}-1}\left[2u\frac{\partial u}{\partial x} + 2v\frac{\partial v}{\partial x}\right]\right]$$

$$= p(u^2 + v^2)^{\frac{p}{2}-1}[uu_{xx} + u_xu_x + vv_{xx} + v_xv_x]$$

$$+p\left(\frac{p}{2} - 1\right)(u^2 + v^2)^{\frac{p}{2}-2}(uu_x + vv_x)(2uu_x + 2vv_x)$$

$$= p(u^2 + v^2)^{\frac{p}{2}-1}[uu_{xx} + u_x^2 + vv_{xx} + v_x^2]$$

$$+2p\left(\frac{p}{2} - 1\right)(u^2 + v^2)^{\frac{p}{2}-2}(uu_x + vv_x)^2$$

Similarly, $\frac{\partial^2}{\partial y^2}(u^2 + v^2)^{p/2} = p(u^2 + v^2)^{\frac{p}{2}-1}\left[uu_{yy} + u_y^2 + vv_{yy} + v_y^2\right]$

$$+2p\left(\frac{p}{2} - 1\right)(u^2 + v^2)^{\frac{p}{2}-2}(uu_y + vv_y)^2$$

$$\Rightarrow \nabla^2|f(z)|^p = p(u^2 + v^2)^{\frac{p}{2}-1}\left[u(u_{xx} + u_{yy}) + v(v_{vx} + v_{yy}) + u_x^2 + u_y^2 + v_x^2 + v_y^2\right] +$$

$$2p\left(\frac{p}{2} - 1\right)(u^2 + v^2)^{\frac{p}{2}-2}\left[u^2u_x^2 + v^2v_x^2 + 2uv\, u_xv_x + u^2u_y^2 + v^2v_y^2 + 2uv\, u_yv_y\right]$$

$$= p(u^2 + v^2)^{\frac{p}{2}-1}\left[u(0) + v(0) + 2(u_x^2 + u_y^2)\right]$$

$$+2p\left(\frac{p}{2}-1\right)(u^2+v^2)^{\frac{p}{2}-2}\left[u^2(u_x^2+u_y^2)+v^2(v_x^2+v_y^2)+2uv(u_xv_x+u_yv_y)\right]$$

$$= 2p(u^2+v^2)^{\frac{p}{2}-1}|f'(z)|^2+2p\left(\frac{p}{2}-1\right)(u^2+v^2)^{\frac{p}{2}-2}[u^2|f'(z)|^2+v^2|f'(z)|^2+2uv(0)]$$

$$= 2p(u^2+v^2)^{\frac{p}{2}-1}|f'(z)|^2+2p\left(\frac{p}{2}-1\right)(u^2+v^2)^{\frac{p}{2}-2}(u^2+v^2)|f'(z)|^2$$

$$= 2p(u^2+v^2)^{\frac{p}{2}-1}|f'(z)|^2+2p\left(\frac{p}{2}-1\right)(u^2+v^2)^{\frac{p}{2}-1}|f'(z)|^2$$

$$= 2p(u^2+v^2)^{\frac{p}{2}-1}|f'(z)|^2\left[1+\frac{p}{2}-1\right]$$

$$= 2p(u^2+v^2)^{\frac{p}{2}-1}|f'(z)|^2 \quad = p^2(u^2+v^2)^{\frac{p-2}{2}}|f'(z)|^2$$

$$= p^2(\sqrt{u^2+v^2})^{p-2}|f'(z)|^2$$

$= p^2|f(z)|^{p-2}|f'(z)|^2$ by (a) & (b)

Theorem: 5 If $\mathbf{f(z) = u + iv}$ is a regular function of z, in a domain D, then

$$\left[\frac{\partial}{\partial x}|f(z)|\right]^2+\left[\frac{\partial}{\partial y}|f(z)|\right]^2=|f'(z)|^2$$

Solution:

Given $f(z) = u + iv$

$$|f(z)| = \sqrt{u^2+v^2}$$

$$\frac{\partial}{\partial x}|f(z)| = \frac{\partial}{\partial x}\left[\sqrt{u^2+v^2}\right]$$

$$= \frac{1}{2\sqrt{u^2+v^2}}[2uu_x+2vv_x] \quad = \frac{uu_x+vv_x}{\sqrt{u^2+v^2}}$$

$$\left[\frac{\partial}{\partial x}|f(z)|\right]^2 = \frac{(uu_x+vv_x)^2}{u^2+v^2} = \frac{u^2u_x^2+v^2v_x^2+2uv\,u_xv_x}{u^2+v^2}$$

Similarly, $\left[\frac{\partial}{\partial y}|f(z)|\right]^2 = \frac{u^2u_y^2+v^2v_y^2+2uv\,u_yv_y}{u^2+v^2}$

$$\left[\frac{\partial}{\partial x}|f(z)|\right]^2+\left[\frac{\partial}{\partial y}|f(z)|\right]^2 = \frac{u^2[u_x^2+u_y^2]+v^2[v_x^2+v_y^2]+2uv\,[u_xv_x+u_yv_y]}{u^2+v^2}$$

$$= \frac{u^2|f'(z)|^2+v^2|f'(z)|^2+2u\,v\,(0)}{u^2+v^2}\,[\because u_x = v_y;\ u_y = -v_x]$$

$$= \frac{(u^2+v^2)|f(z)|^2}{u^2+v^2} = |f'(z)|^2[\because u_xv_x+u_yv_y = 0]$$

Theorem: 6 If $\mathbf{f(z)} = \boldsymbol{u+iv}$ is a regular function of z, then $\mathbf{\nabla^2|Re\ f(z)|^2 = 2|f'(z)|^2}$

Solution:

Let $f(z) = u + iv$

$\mathrm{Re} f(z) = u$

$|\mathrm{Re}\, f'(z)|^2 = u^2$

$\nabla^2|\mathrm{Re}\, f'(z)|^2 = \nabla^2 u^2$

$$= \left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)(u^2)$$

$$= \left(\frac{\partial^2}{\partial x^2}\right)(u^2) + \left(\frac{\partial^2}{\partial y^2}\right)(u^2)$$

$$= 2[u_x^2+u_y^2]$$

$$= 2\,|f'(z)|^2$$

Theorem: 7 If $\mathbf{f(z)} = \boldsymbol{u + iv}$ is a regular function of z, then prove that $\boldsymbol{\nabla^2|\mathrm{Im}\,f(z)|^2 = 2|f'(z)|^2}$

Proof:

Let $f(z) = u + iv$

$\mathrm{I\,m}f(z) = v$

$|Im\,f(z)|^2 = v^2$

$$\frac{\partial}{\partial x}(v^2) = 2vv_x$$

$$\frac{\partial^2}{\partial x^2}(v^2) = 2[vv_{xx} + v_x v_x] = 2[vv_{xx} + v_x^2]$$

Similarly, $\frac{\partial^2}{\partial y^2}(v^2) = 2[vv_{yy} + v_y^2]$

$$\therefore \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|Im\,f(z)|^2 = 2[v(v_{xx} + v_{yy}) + v_x^2 + v_y^2]$$

$$= 2[v(0) + u_x^2 + v_x^2] \qquad \text{by C–R equation}$$

$$= 2|\mathrm{f}'(\mathrm{z})|^2$$

Theorem: 8 Show that $\boldsymbol{\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4\frac{\partial^2}{\partial z\partial \bar{z}}}$ (or) S T $\nabla^2 = \boldsymbol{4\frac{\partial^2}{\partial z\partial \bar{z}}}$

Proof:

Let x & y are functions of z and $\bar{z}$

that is $x = \frac{z+\bar{z}}{2}, y = \frac{z-\bar{z}}{2i}$

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial x}\frac{\partial x}{\partial z} + \frac{\partial}{\partial y}\frac{\partial y}{\partial z}$$

$$= \frac{\partial}{\partial x}\left(\frac{1}{2}\right) + \frac{\partial}{\partial y}\left[\frac{1}{2i}\right] = \frac{1}{2}\left[\frac{\partial}{\partial x} + \frac{1}{i}\frac{\partial}{\partial y}\right]$$

$$2\frac{\partial}{\partial z} = \frac{\partial}{\partial x} + \frac{1}{i}\frac{\partial}{\partial y} \qquad \ldots(1)$$

$$\frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial x}\frac{\partial x}{\partial z} + \frac{\partial}{\partial y}\frac{\partial y}{\partial \bar{z}}$$

$$= \frac{\partial}{\partial x}\left(\frac{1}{2}\right) + \frac{\partial}{\partial y}\left[\frac{-1}{2i}\right] = \frac{1}{2}\left[\frac{\partial}{\partial x} - \frac{1}{i}\frac{\partial}{\partial y}\right]$$

$$2\frac{\partial}{\partial \bar{z}} = \left(\frac{\partial}{\partial x} - \frac{1}{i}\frac{\partial}{\partial y}\right) \qquad \ldots(2)$$

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \left(\frac{\partial}{\partial x} + \frac{1}{i}\frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x} - \frac{1}{i}\frac{\partial}{\partial y}\right) [\because (a+b)(a-b) = a^2 - b^2]$$

$$= \left(2\frac{\partial}{\partial z}\right)\left(2\frac{\partial}{\partial \bar{z}}\right) \text{ by (1) \& (2)}$$

$$= 4\frac{\partial^2}{\partial z\partial \bar{z}}$$

Theorem: 9 If $\boldsymbol{f(z)}$ is analytic, show that $\boldsymbol{\nabla^2|f(z)|^2 = 4|f'(z)|^2}$

Solution:

We know that, $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4\frac{\partial^2}{\partial z\partial \bar{z}}$

$$|f(z)|^2 = f(z)\overline{f(z)}$$

$$\nabla^2 |\, f(z)|^2 = 4\frac{\partial}{\partial z}\frac{\partial}{\partial \bar{z}}[f(z)\overline{f(z)}]$$

$$= 4\left[\frac{\partial}{\partial z}f(z)\right]\left[\frac{\partial}{\partial \bar{z}}\overline{f(z)}\right]$$

[$\because f(z)$is independent of $\bar{z}$and $\overline{f(z)}$ is independent of z]

$$\therefore\ \nabla^2 |\, f(z)|^2 = 4[f'(z)\left[\overline{\frac{\partial}{\partial z}f(z)}\right] = 4f'(z)\overline{f'(z)}$$

$$= 4|\, f'(z)|^2 \quad [\because z\bar{z} = |\,(z)|^2]$$

Example: 3.20 Give an example such that u and v are harmonic but $u + iv$ is not analytic.

Solution:

$$u = x^2 - y^2,\ \ v = \frac{-y}{x^2+y^2}$$

Example: 3.21 Find the value of m if $u = 2x^2 - my^2 + 3x$ is harmonic.

Solution:

Given $u = 2x^2 - my^2 + 3x$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \text{ [}\because \text{ u is harmonic]} \qquad \ldots(1)$$

$$\frac{\partial u}{\partial x} = 4x + 3 \qquad \frac{\partial u}{\partial y} = -2my$$

$$\frac{\partial^2 u}{\partial x^2} = 4 \qquad \frac{\partial^2 u}{\partial y^2} = -2m$$

$$\therefore (1) \Rightarrow (4) + (-2m) = 0$$

$$\Rightarrow m = 2$$

3.4 CONSTRUCTION OF ANALYTIC FUNCTION

There are three methods to find $f(z)$.

Method: 1 Exact differential method.

(i) Suppose the harmonic function $u(x, y)$ is given.

Now, $dv = v_x dx + v_y dy$ is an exact differential

Where, v_x and v_y are known from u by using C−R equations.

$$\therefore v = \int v_x\, dx + \int v_y dy = -\int u_y dx + \int u_x dy$$

(ii) Suppose the harmonic function $v(x, y)$ is given.

Now, $du = u_x dx + u_y dy$ is an exact differential

Where, u_x and u_y are known from v by using C−R equations.

$$u = \int u_x\, dx + \int u dy$$

$$= \int v_y dx + \int -v_x dy$$

$$= \int v_y dx - \int v_x dy$$

Method: 2 Substitution method

$$f(z) = 2u\left[\frac{1}{2}(z+a), \frac{-i}{2}(z-a)\right] - [u(a,0), -iv(a,0)]$$

Here, $u(a,0), -iv(a,0)$ is a constant

Thus $f(z) = 2u\left[\frac{1}{2}(z+a), \frac{-i}{2}(z-a)\right] + C$

By taking $a = 0$, that is, if $f(z)$ is analytic $z = 0 + i0$,

We have the simpler formula for $f(z)$

$$f(z) = 2\left[u\frac{z}{2}, \frac{-iz}{2}\right] + C$$

Method: 3 [Milne – Thomson method]

(i) $\boxed{\text{To find } f(z) \text{ when } u \text{ is given}}$

Let $f(z) = u + iv$

$f'(z) = u_x + iv_x$

$= u_x - iv_y$ [by C−R condition]

$\therefore f(z) = \int u_x(z,0)dz - i\int u_y(z,0)dz + C$ [by Milne−Thomson rule],

Where, C is a complex constant.

(ii) $\boxed{\text{To find } f(z) \text{ when v is given}}$

Let $f(z) = u + iv$

$f'(z) = u_x + iv_x$

$= v_y + iv_x$ [by C−R condition]

$\therefore f(z) = \int v_y(z,0)dz + i\int v_x(z,0)dz + C$ [by Milne−Thomson rule],

Where, C is a complex constant.

Example: 3.22 Construct the analytic function $f(z)$ for which the real part is $e^x \cos y$.

Solution:

Given $u = e^x \cos y$

$\Rightarrow u_x = e^x \cos y$ $[\therefore \cos 0 = 1]$

$\Rightarrow u_x(z,0) = e^x$

$\Rightarrow u_y = e^x \cos y$ $[\therefore \sin 0 = 0]$

$\Rightarrow u_y(z,0) = 0$

$\therefore f(z) = \int u_x(z,0)dz - i\int u_y(z,0)dz + C$ [by Milne−Thomson rule],

Where, C is a complex constant.

$\therefore f(z) = \int e^z\, dz - i\int 0\, dz + C$

$= e^z + C$

Example: 3.23 Determine the analytic function $w = u + iv$ if $u = e^{2x}(x\cos 2y - y\sin 2y)$

Solution:

Given $u = e^{2x}(x\cos 2y - y\sin 2y)$

$u_x = e^{2x}[\cos 2y] + (x cos2y - y\sin 2y)[2\, e^{2x}]$

$$u_x(z,0) = e^{2z}[1]+[z(1)-0][2e^{2z}]$$
$$= e^{2z} + 2ze^{2z}$$
$$= (1+2z)e^{2x}$$
$$u_y = e^{2x}[-2x\sin 2y - (y2cos2y + \sin 2y)]$$
$$u_y(z,0) = e^{2z}[-0-(0+0)] = 0$$

$\therefore f(z) = \int u_x(z,0)dz - i\int u_y(z,0)dz + C$ [by Milne−Thomson rule],

Where, C is a complex constant.

$$f(z) = \int (1+2z)e^{2z}dz - i\int 0 + dz + C$$
$$= \int (1+2z)e^{2z}dz + C$$
$$= (1+2z)\frac{e^{2z}}{2} - 2\frac{e^{2z}}{4} + C \quad [\because \int uv\,dz = uv_1 - u'v_2 + u''v_3 - \cdots]$$
$$= \frac{e^{2z}}{2} + ze^{2z} - \frac{e^{2z}}{2} + C$$
$$= ze^{2z} + C$$

Example: 3.24 Determine the analytic function where real part is

$\boldsymbol{u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1}$.

Solution:

Given $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$

$$u_x = 3x^2 - 3y^2 + 6x$$
$$\Rightarrow u_x(z,0) = 3z^2 - 0 + 6z$$
$$u_y = 0 - 6xy + 0 - 6y$$
$$\Rightarrow u_y(z,0) = 0$$

$f(z) = \int u_x(z,0)dz - i\int u_y(z,0)dz + C$ [by Milne−Thomson rule],

Where, C is a complex constant.

$$f(z) = \int (3z^2 + 6z)dz - i\int 0 + dz + C$$
$$= 3\frac{z^2}{3} + 6\frac{z^2}{2} + C$$
$$= z^3 + 3z^2 + C$$

Example: 3.25 Determine the analytic function whose real part in $\boldsymbol{\frac{\sin 2x}{\cosh 2y - \cos 2x}}$

Solution:

Given $u = \frac{\sin 2x}{\cosh 2y - \cos 2x}$

$$u_x = \frac{(\cosh 2y - \cos 2x)[2\cos 2x] - \sin 2x[2\sin 2x]}{[\cosh 2y - \cos 2x]^2}$$
$$u_x(z,0) = \frac{(1-\cos 2z)(2\cos 2z) - 2\sin^2 2z}{[\cosh 0 - \cos 2z]^2}$$
$$= \frac{2\cos 2z - 2\cos^2 2z - 2\sin^2 2z}{(1-\cos 2z)^2}$$

$$= \frac{2\cos 2z - 2[\cos^2 2z + \sin^2 2z]}{(1-\cos 2z)^2}$$

$$= \frac{2\cos 2z - 2}{(1-\cos 2z)^2}$$

$$= \frac{-2(1-\cos 2z)}{(1-\cos 2z)^2}$$

$$= \frac{2\cos 2z - 2}{(1-\cos 2z)}$$

$$= \frac{-2}{2sin^2 2}$$

$$= -cosec^2 z$$

$$u_y = \frac{(\cosh 2y - \cos 2x)\,(0) - \sin 2x[2\sin 2y]}{[\cosh 2y - \cos 2x]^2}$$

$$\Rightarrow\ u_y(z,0) = 0$$

$f(z) = \int u_x\,(z,0)dz - i\int u_y(z,0)dz + C$ [by Milne−Thomson rule],

where C is a complex constant.

$$f(z) = \int(-cosec^2 z)dz - i\int 0\,dz + C$$

$$= cot\ z + C$$

Example: 3.26 Show that the function $u = \frac{1}{2}\log(x^2 + y^2)$ is harmonic and determine its conjugate.

Also find $f(z)$

Solution:

Given $u = \frac{1}{2}\log(x^2 + y^2)$

$$u_x = \frac{1}{2}\frac{1}{(x^2+y^2)}(2x) = \frac{x}{x^2+y^2},$$

$$\Rightarrow u_x(z,0) = \frac{z}{z^2} = \frac{1}{z}$$

$$u_{xx} = \frac{(x^2+y^2)[1] - x[2x]}{[x^2+y^2]^2} = \frac{x^2+y^2-2x^2}{[x^2+y^2]^2} = \frac{y^2-x^2}{[x^2+y^2]^2} \quad \ldots(1)$$

$$u_y = \frac{1}{2}\frac{1}{x^2+y^2}(2y) = \frac{y}{x^2+y^2}$$

$$\Rightarrow u_y(z,0) = 0$$

$$u_{yy} = \frac{(x^2+y^2)[1] - y[2y]}{[x^2+y^2]^2} = \frac{x^2-y^2}{[x^2+y^2]^2} \quad \ldots(2)$$

To prove u is harmonic:

$$\therefore u_{xx} + u_{yy} = \frac{(y^2-x^2)+(x^2-y^2)}{[x^2+y^2]^2} = 0 \qquad by\ (1)\&(2)$$

$\Rightarrow u$ is harmonic.

To find $f(z)$:

$f(z) = \int u_x\,(z,0)dz - i\int u_y(z,0)dz + C$ [by Milne−Thomson rule],

Where, C is a complex constant.

$$f(z) = \int \frac{1}{z}dz - i\int 0\,dz + C$$

$$= log\ z + C$$

To find v :

$$f(z) = log\,(re^{i\theta}) \qquad [\because z = re^{i\theta}]$$

$$u + iv = \log r + \log\, e^{i\theta} = \log r + i\theta$$

$$\Rightarrow u = logr, v = \theta$$

Note: $z = x + iy$

$$r = |z| = \sqrt{x^2 + y^2}$$

$$\log r = \frac{1}{2}\log(\,x^2 + y^2)$$

$$\tan\theta = \frac{y}{x}$$

$$\theta = tan^{-1}\left(\frac{y}{x}\right) \qquad i.e., v = tan^{-1}\left(\frac{y}{x}\right)$$

Example: 3.27 Construct an analytic function $f(z) = u + iv$, given that $u = e^{x^2-y^2}\cos 2xy$. Hence find v.

Solution:

Given $u = e^{x^2-y^2}\cos 2xy = e^{x^2}e^{-y^2}\cos 2xy$

$$u_x = e^{-y^2}\left[e^{x^2}(-2y\sin 2xy) + \cos 2xy\; e^{x^2}2x\right]$$

$$u_x(z,0) = 1\left[e^{z^2}(0) + 2ze^{z^2}\right] = 2ze^{z^2}$$

$$u_y = e^{x^2}\left[e^{-y^2}(-2x\sin 2xy) + \cos 2xye^{-y^2}(-2y)\right]$$

$$u_y(z,0) = e^{z^2}[0+0] = 0$$

$$f(z) = \int u_x\,(z,0)dz - i\int u_y(z,0)dz + C \quad \text{[by Milne–Thomson rule]}$$

$$= \int 2z\,e^{z^2}dz + C$$

$$= 2\int z\,e^{z^2}dz + C$$

put $t = z^2$, dt$= 2z\,dz$

$$= \int e^t\,dt + C$$

$$= e^t + C$$

$$f(z) = e^{z^2} + C$$

To find v :

$$u + iv = e^{(x+iy)^2} = e^{x^2-y^2+i\,2xy} = e^{x^2-y^2}e^{i2\,xy}$$

$$= e^{x^2-y^2}\,[\cos(2xy) + isin\,(2xy)]$$

$$v = e^{x^2-y^2}\sin 2xy \qquad [\because\text{equating the imaginary parts}]$$

Example: 3.28 Find the regular function whose imaginary part is $e^{-x}(x\cos y + y\sin y)$.

Solution:

Given $v = e^{-x}(x\cos y + y\sin y)$

$$v_x = e^{-x}[\cos y] + (x\cos y + y\sin y)[-e^{-x}]$$

$$v_x(z,0) = e^{-z} + (z)(-e^{-z}) = (1-z)e^{-z}$$

$$v_y = e^{-x}[-x\sin y + (y\cos y \ + \sin y\,(1))]$$

$$v_x(z,0) = e^{-z}[0+0+0] = 0$$

$\therefore f(z) = \int v_y\,(z,0)dz + i\int v_x(z,0)dz + C$ [by Milne−Thomson rule]

Where, C is a complex constant.

$$f(z) = \int 0dz + i\int (1-z)e^{-z}dz + C$$

$$= i\int (1-z)e^{-z}dz + C$$

$$= i\left[(1-z)\left[\frac{e^{-z}}{-1}\right] - (-1)\left[\frac{e^{-z}}{(-1)^2}\right]\right] + C$$

$$= i[-(1-z)e^{-z} + e^{-z}] + C$$

$$= ize^{-z} + C$$

Example: 3.29 In a two dimensional flow, the stream function is $\psi = tan^{-1}\left(\frac{y}{x}\right)$. Find the velocity potential $\boldsymbol{\varphi}$.

Solution:

Given $\psi = tan^{-1}(y/x)$

We should denote, ϕ by u and $\psi$by v

$$\therefore v = tan^{-1}(y/x)$$

$v_x = \frac{1}{1+(y/x)^2}\left[\frac{-y}{x^2}\right] = \frac{-y}{x^2+y^2}$, $v_x(z,0) = 0$

$v_y = \frac{1}{1+(y/x)^2}\left[\frac{1}{x}\right] = \frac{x}{x^2+y^2}$ $v_x(z,0) = \frac{z}{z^2} = \frac{1}{z}$

$$\therefore f(z) = \int v_y\,(z,0)dz + i\int v_x(z,0)dz + C$$

$$f(z) = \int \frac{1}{z}dz + i\int 0\,dz + C \ = \log z + C$$

To find $\boldsymbol{\varphi}$:

$f(z) = log\,(re^{i\theta})$ $[\because z = re^{i\theta}]$

$$u + iv = \log r + \log\, e^{i\theta}$$

$$u + iv = \log r + i\theta$$

$\Rightarrow u = \log r$ $\boxed{z = x + iy, |z| = \sqrt{x^2+y^2}}$

$$\Rightarrow\ u = log\sqrt{x^2+y^2}$$

$$= \frac{1}{2}\log(x^2+y^2)$$

So, the velocity potential φ is

$$\varphi = \frac{1}{2}\log(x^2+y^2)$$

Note: In two dimensional steady state flows, the complex potential $f(z) = \varphi(x,y) + i\psi(x,y)$ is analytic.

Example: 3.30 If $\boldsymbol{w = u + iv}$ is an analytic function and $\boldsymbol{v = x^2 - y^2 + \frac{x}{x^2+y^2}}$, find $\boldsymbol{u}$.

Solution:

Given $v = x^2 - y^2 + \frac{x}{x^2+y^2}$

$$v_x = 2x - 0 + \frac{(x^2+y^2)(1) - x(2x)}{(x^2+y^2)^2}$$

$$= 2x + \frac{y^2-x^2}{(x^2+y^2)^2}, \qquad v_x(z,0) = 2z + \frac{(-z^2)}{(z^2)}$$

$$\Rightarrow v_x(z,0) = 2z - \frac{1}{z^2}$$

$$v_y = 0 - 2y + \frac{0 - x(2y)}{(x^2+y^2)^2}$$

$$= 0 - 2y - \frac{2xy}{(x^2+y^2)^2}$$

$$\Rightarrow v_y(z,0) = 0$$

$\therefore f(z) = \int v_y\,(z,0)dz + i\int v_x(z,0)dz + C$ [by Milne−Thomson rule]

Where, C is a complex constant.

$$f(z) = \int 0dz + i\int \left(2z - \frac{1}{z^2}\right)dz + C$$

$$= i\left[2\frac{z^2}{2} + \frac{1}{z}\right] + C \qquad \left[\because \int \frac{-1}{z^2}dz = \frac{1}{z}\right]$$

$$= i\left[z^2 + \frac{1}{z}\right] + C$$

Example: 3.31 If $f(z) = u + iv$ is an analytic function and $u - v = e^x(\cos y - \sin y)$, find $f(z)$ interms of z.

Solution:

Given $u - v = e^x(\cos y - \sin y)$, $\quad \ldots(A)$

Differentiate (A) p.w.r. to x, we get

$$u_x - v_x = e^x(\cos y - \sin y),$$

$$u_x(z,0) - v_x(z,0) = e^z \qquad \ldots(1)$$

Differentiate (A) p.w.r. to y, we get

$$u_y - v_y = e^x(-\sin y - \cos y)$$

$$u_y(z,0) - v_y(z,0) = e^z[-1]$$

$$\text{i.e., } u_y(z,0) - v_y(z,0) = -e^z$$

$-v_x(z,0) - u_x(z,0) = -e^z$ $\quad \ldots(2)$ [by C−R conditions]

$$(1) + (2) \Rightarrow -2v_x(z,0) = 0$$

$$\Rightarrow v_x(z,0) = 0$$

$$(1) \Rightarrow u_x(z,0) = e^z$$

$f(z) = \int u_x\,(z,0)dz + i\int v_x(z,0)dz + C$ [by Milne−Thomson rule]

$$f(z) = \int e^z dz + i0 + C$$

$$= e^z + C$$

Example: 3.32 Find the analytic functions $f(z) = u + iv$ given that

(i) $\boldsymbol{2u + v = e^x(\cos y - \sin y)}$

(ii) $\boldsymbol{u - 2v = e^x(\cos y - \sin y)}$

Solution:

Given (i) $2u + v = e^x(\cos y - \sin y) \qquad \dots(A)$

Differentiate (A) p.w.r. to x, we get

$$2u_x + v_x = e^x(\cos y - \sin y)$$

$$2u_x - u_y = e^x(\cos y - \sin y) \quad \text{[by C−R condition]}$$

$$2u_x(z,0) - u_y(z,0) = e^z \qquad \dots(1)$$

Differentiate (A) p.w.r. to y, we get

$$2u_y + v_y = e^x[-\sin y - \cos y]$$

$$2u_y + u_x = e^x\,[-\sin y - \cos y] \quad \text{[by C−R condition]}$$

$$2u_y(z,0) + u_x(z,0) = e^z(-1) = -e^z \qquad \dots(2)$$

$(1) \times (2) \quad \Rightarrow 4u_x(z,0) - 2u_y(z,0) = 2e^z \qquad \dots(3)$

$(2) + (3) \quad \Rightarrow 5u_x(z,0) = e^z$

$\Rightarrow u_x(z,0) = \frac{1}{5}e^z$

$(1) \quad \Rightarrow u_y(z,0) = \frac{2}{5}e^z - e^z = -\frac{3}{5}e^z$

$\Rightarrow u_y(z,0) = -\frac{3}{5}e^z$

$f(z) = \int u_x\,(z,0)dz - i\int u_y(z,0)dz + C$ [by Milne−Thomson rule]

Where, C is a complex constant.

$$f(z) = \int \frac{1}{5}e^z dz - i\int -\frac{3}{5}e^z dz + C$$

$$= \frac{2}{5}e^z + \frac{3}{5}ie^z + C$$

$$= \frac{1+3i}{5}e^z + C$$

(ii) $u - 2v = e^x(\cos y - \sin y) \qquad \dots(B)$

Differrentiate (B) p.w.r. to x, we get

$$u_x - 2v_x = e^x(\cos y - \sin y)$$

$$u_x + 2u_y = e^x(\cos y - \sin y) \quad \text{[by C−R condition]}$$

$$u_x(z,0) + 2u_y(z,0) = e^z \qquad \dots(1)$$

Differentiate (B) p.w.r. to y, we get

$$u_y - 2v_y = e^x[-\sin y - \cos y]$$

$$u_y - 2u_x = e^x\,[-\sin y - \cos y] \quad \text{[by C−R condition]}$$

$$u_y(z,0) - 2u_x(z,0) = -e^z \qquad \dots(2)$$

$(1) \times (2) \quad \Rightarrow 2u_x(z,0) + 4u_y(z,0) = 2e^z \qquad \dots(3)$

$(2) + (3) \quad \Rightarrow 5u_y(z,0) = e^z$

$$\Rightarrow u_y(z,0) = \frac{1}{5}e^z$$

(1) $\Rightarrow u_x(z,0) = -\frac{2}{5}e^z + e^z$

$$= \frac{3}{5}e^z$$

$f(z) = \int u_x\,(z,0)dz - i\int u_y(z,0)dz + C$ [by Milne−Thomson rule]

Where, C is a complex constant.

$$f(z) = \int \frac{3}{5}e^z dz - i\int \frac{1}{5}e^z dz + C$$

$$= \frac{3}{5}e^z - i\frac{1}{5}e^z + C = \frac{3-i}{5}e^z + C$$

Example: 3.33 Determine the analytic function $f(z) = u + iv$ given that $3u + 2v = y^2 - x^2 + 16xy$

Solution:

Given $3u + 2v = y^2 - x^2 + 16xy$ $\ldots(A)$

Differentiate (A) p.w.r. to x, we get

$$3u_x + 2v_x = -2x + 16y$$

$3u_x - 2u_y = -2x + 16y$ [by C−R condition]

$$3u_x(z,0) - 2u_y(z,0) = -2z \quad \ldots(1)$$

Differentiate (A) p.w.r. to y, we get

$$3u_y + 2v_y = 2y + 16x$$

$3u_y + 2u_x = 2y + 16x$ [by C−R condition]

$$3u_y(z,0) + 2u_x(z,0) = 16z \quad \ldots(2)$$

$(1)\times(2)$ $\Rightarrow 6u_x(z,0) - 4u_y(z,0) = -4z$ $\ldots(3)$

$(2)\times(3)$ $\Rightarrow 9u_y(z,0) + 6u_x(z,0) = 48z$

$(3)-(4)$ $\Rightarrow -13u_y(z,0) = -52z$

$$\Rightarrow u_y(z,0) = 4z$$

(1) $\Rightarrow 3u_x(z,0) = 8z - 2z = 6z$

$$\Rightarrow u_x(z,0) = 2z$$

$f(z) = \int u_x\,(z,0)dz - i\int u_y(z,0)dz + C$ [by Milne−Thomson rule]

where C is a complex constant.

$$f(z) = \int 2z dz - i\int 4z dz + C$$

$$= 2\,\frac{z^2}{2} - i\frac{4z^2}{2} + C$$

$$= z^2 - i2z^2 + C$$

$$= (1 - i2)z^2 + C$$

Example:3.34 Find an analytic function $f(z) = u + iv$ given that $2u + 3v = \frac{\sin 2x}{\cosh 2y - \cos 2x}$

Solution:

Given $2u + 3v = \frac{\sin 2x}{\cosh 2y - \cos 2x}$

Differentiate p.w.r. to x, we get

$$2u_x + 3v_x = \frac{(\cosh 2y - \cos 2x)(2\cos 2x) - \sin 2x\,(2\sin 2x)}{(\cosh 2y - \cos 2x)^2}$$

$$2u_x - 3u_y = \frac{(\cosh 2y - \cos 2x)(2\cos 2x) - \sin 2x\,(2\sin 2x)}{(\cosh 2y - \cos 2x)^2} \quad \text{[by C−R condition]}$$

$$2u_x(z,0) - 3u_y(z,0) = \frac{2\cos 2z(1 - \cos 2z) - 2\sin^2 2z}{(1 - \cos 2z)^2}$$

$$= \frac{2\cos 2z - 2\cos^2 2z - 2\sin^2 2z}{(1 - \cos 2z)^2}$$

$$= \frac{2\cos 2z - 2}{(1 - \cos 2z)^2} = \frac{-2}{1 - \cos 2z}$$

$$= \frac{-2}{2\sin^2 z} = -cosec^2 z$$

$$2u_x(z,0) - 3u_y(z,0) = -cosec^2 z \qquad \ldots(1)$$

Differentiate p.w.r. to y, we get

$$2u_y + 3v_y = \frac{0 - \sin 2x(\sinh 2y)}{(\cosh 2y - \cos 2x)^2}(2)$$

$$2u_y + 3u_x = \frac{0 - \sin 2x(\sinh 2y)}{(\cosh 2y - \cos 2x)^2}(2) \qquad \text{[by C − R condition]}$$

$$2u_y(z,0) + 3u_x(z,0) = 0 \qquad \ldots(2)$$

Solving (1) & (2) we get,

$$\Rightarrow u_x(z,0) = -\frac{2}{13}\text{cosec}^2 z$$

$$\Rightarrow u_y(z,0) = -\frac{2}{13}\text{cosec}^2 z$$

$$f(z) = \int u_x\,(z,0)dz - i\int u_y(z,0)dz + C \quad \text{[by Milne−Thomson rule]}$$

Where, C is a complex constant

$$f(z) = \int \left(\frac{-2}{13}\right)\text{cosec}^2 z\,dz - i\int \left(\frac{3}{13}\right)\text{cosec}^2 z dz + C$$

$$= \left(\frac{2}{13}\right)\cot z + \left(\frac{3}{13}\right)\cot z + C$$

$$= \frac{2+3i}{i3}\cot z + C$$

Example: 3.35 Find the analytic function $f(z) = u + iv$ given that $2u + 3v = e^x(\cos y - \sin y)$

Solution:

Given $2u + 3v = e^x(\cos y - \sin y)$

Differentiate p.w.r. to x, we get

$$2u_x + 3v_x = e^x(\cos y - \sin y)$$

$$2u_x - 3u_y = e^x(\cos y - \sin y) \qquad \text{[by C−R condition]}$$

$$2u_x(z,0) - 3u_y(z,0) = e^z \qquad \ldots(1)$$

Differentiate p.w.r. to y, we get

$$2u_y + 3v_y = e^x[-\sin y - \cos y]$$

$$2u_y + 3u_x = -e^x\,[\sin y + \cos y] \qquad \text{[by C−R condition]}$$

$$2u_y(z,0) + 3u_x(z,0) = -e^z \qquad \ldots(2)$$

$$(1)\times(3) \quad \Rightarrow 6u_x(z,0) - 9u_y(z,0) = 3e^z \qquad \ldots(3)$$

$$(2)\times 2 \quad \Rightarrow \; 6u_x(z,0) + 4u_y(z,0) = -2e^z \qquad \ldots(4)$$

$$(3)-(4) \;\; \Rightarrow \; -13u_y(z,0) = 5e^z$$

$$\Rightarrow u_y(z,0) = -\frac{5}{13}e^z$$

$$(1) \quad \Rightarrow 2u_x(z,0) + \frac{15}{13}e^z = e^z$$

$$2u_x(z,0) = e^z - \frac{15}{13}e^z = -\frac{2}{13}e^z$$

$$\Rightarrow u_x(z,0) = -\frac{1}{13}e^z$$

$$f(z) = \int u_x\,(z,0)dz - i\int u_y(z,0)dz + C$$

$$\therefore f(z) = \int \frac{-1}{13}e^z\;dz - i\int\left(\frac{-5}{13}\right)dz + C$$

$$= \frac{-1}{13}e^z + \frac{5}{13}e^z i + C = \frac{-1+5i}{13}e^z + C$$

Exercise: 3.4

Construction of an analytic function

1. Show that the function $u(x,y) = 3x^2y + 2x^2 - y^3 - 2y^2$ is harmonic. Find the conjugate harmonic function v and express $u + iv$ as an analytic function of z.

   [**Ans:** $v(x,y) = 3x^2y + 4xy - x^3 + C$, $f(z) = -iz^3 + 2z^2 + iC$ where C is a real constant.]

2. If $f(z) = u + iv$ is an analytic function of z, and if $u = \frac{2\sin 2x}{e^{2y}+e^{-2y}-2\cos 2x}$, find v.

   [**Ans**: $v = \frac{-2\sinh 2y}{e^{2y}+e^{-2y}-2\cos 2x} + C$]

3. Find v such that $w = u + iv$ is an analytic function of z, given that $u = e^{x^2-y^2}\cos 2xy$. Hence find w.

   [**Ans:** $v = e^{x^2-y^2}\sin 2xy + C \;\; w = e^{z^2} + C$]

4. Find the analytic function $w = u + iv$ if $w = e^{2^x}(x\cos 2y - y\sin 2y)$.Hence find u.

   [**Ans**: $w = ize^{2z} + C, u = -(x\sin 2y + y\cos 2y)\,e^{2x} + C$]

5. If $v = \frac{x-y}{x^2+y^2}$ find u such $u + iv$ is an analytic function. What is the harmonic conjugate of v?

   [**Ans**: $u = \frac{x+y}{x^2+y^2} + C$ Harmonic conjugate of v is $-u = \frac{-(x+y)}{x^2+y^2}$, $f(z) = \frac{1+i}{z} + C$

6. Find the analytic function whose real part is $\frac{\sin 2x}{\cosh 2y+\cos 2x}$

   [**Ans:** $f(z) = \tan z + C$]

7. Find the analytic function whose imaginary part is $-e^{-2xy}\cos(x^2-y^2)$

[**Ans:** $f(z)=-ie^{iz^2}+C$]

6. Prove that $u=2^x-x^3+3xy^2$ is harmonic and find its harmonic conjugate. Also find the corresponding analytic function. [**Ans**: $v=2y-3x^2y+y^3+C, f(z)=2z-z^3+iC$]

7. Find the real part of the analytic function whose imaginary part is $e^{-x}[2xy\cos y+(y^2-x^2)\sin y]$. Construct the analytic function.

   [**Ans:** $u=e^{-x}[(x^2-y^2)\cos y+2xy\sin y], f(z)=z^2e^{-z}+C$]

8. Find the analytic function $f(z)=u+iv$ given that $2u+v=e^{2x}[(2x+y)\cos 2y+(x-2y)\sin 2y]$

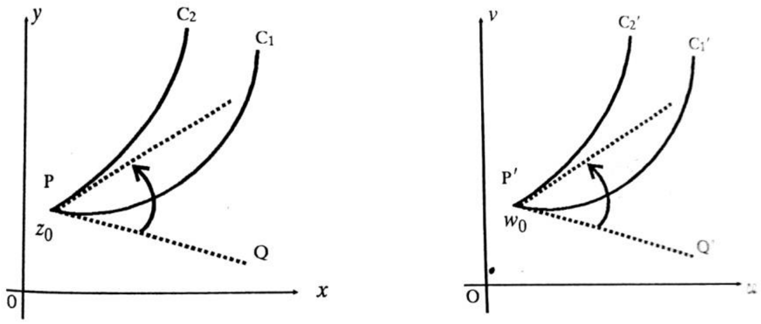
   [**Ans**: $f(z)=ze^{2z}+C$]

9. Prove that $u=x^2-y^2$ and $v=-\frac{y}{x^2+y^2}$ are harmonic functions but not harmonic conjugates.

3.5 CONFORMAL MAPPING

Definition: Conformal Mapping

A transformation that preserves angels between every pair of curves through a point, both in magnitude and sense, is said to be conformal at that point.



Definition: Isogonal

A transformation under which angles between every pair of curves through a point are preserved in magnitude, but altered in sense is said to be an isogonal at that point.

Note: 3.4 (i) A mapping $w=f(z)$ is said to be conformal at $z=z_0$, if $f'(z_0)\neq 0$.

Note: 3.4 (ii) The point, at which the mapping $w=f(z)$ is not conformal,

$(i.e.)f'(z)=0$ is called a **critical point** of the mapping.

If the transformation $w=f(z)$ is conformal at a point, the inverse transformation $z=f^{-1}(w)$ is also conformal at the corresponding point.

The critical points of $z=f^{-1}(w)$ are given by $\frac{dz}{dw}=0$. hence the critical point of the transformation $w=f(z)$ are given by $\frac{dw}{dz}=0$ and $\frac{dz}{dw}=0$,

Note: 3.4 (iii) Fixed points of mapping.

Fixed or invariant point of a mapping $w=f(z)$ are points that are mapped onto themselves, are "Kept fixed" under the mapping. Thus they are obtained from $w=f(z)=z$.